

Quantum alternating operator ansatz with PSO optimizer for portfolio optimization problem

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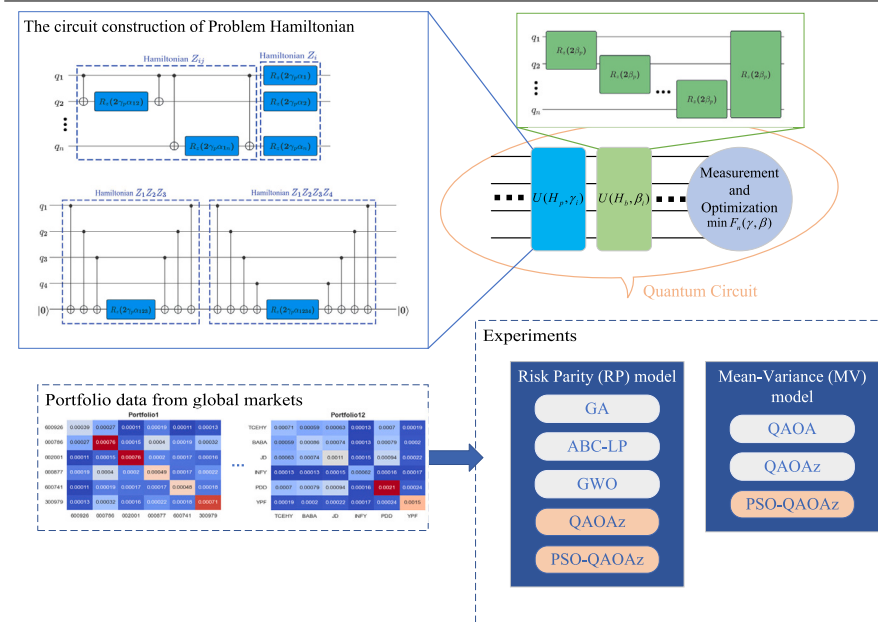
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GRAPHICAL ABSTRACT



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ABSTRACT

Recently, Quantum Approximate Optimization Algorithm (QAOA) and Quantum Alternating Operator ansatz (QAOAz) are utilized to solve the Mean Variance (MV) model for the portfolio optimization problem (QAOAz-MV), which shows performance advantages over classical algorithms on this huge search space. For the more complex and comprehensive risk parity (RP) model, a novel QAOAz solution (QAOAz-RP) is proposed. We begin by defining the RP model for the portfolio optimization problem. Next, we detail the QAOAz algorithm process, where the problem Hamiltonian with the ZZZZ term is derived, the corresponding quantum circuit is

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ingeniously constructed using the parity check method, and the whole quantum circuit containing the ring XY-mixer is given. Finally, to improve the optimization performance of QAOAz, a Particle Swarm Optimization (PSO) optimizer is introduced to tune the parameters of the quantum circuits, which is applicable to both QAOAz-MV and QAOAz-RP. The experiment conducted on multiple financial markets (e.g. Chinese, U.S., and European) demonstrate that PSO-QAOAz-RP is significantly better for portfolio optimization than ABC-LP, GWO, GA and QAOAz-RP on eight portfolios in all metrics. PSO-QAOAz-MV also has advantage for MV model over the quantum algorithms, including QAOA (improves approximate ratio by 54.79% on average) and QAOAz (improves approximate ratio by 15.38% on average). This study not only provides a breakthrough quantum solution for portfolio optimization, but also provides a reusable technology paradigm for the deep integration of quantum computing and financial engineering.

1. Introduction

Portfolio optimization, as a core issue in the financial sector, has received extensive development and application in both theory and practice [1]. Since Markowitz proposed the mean variance (MV) model in 1952 [2], portfolio theory has rapidly evolved, including the development of the risk parity (RP) model [3] emerging in response to these advancements. These research achievements not only provide investors with scientific and rational theoretical support for investment decision-making but also make a positive contribution to the stability and development of financial markets. However, portfolio optimization still faces many challenges, such as the complexity of investment strategies [4,5]. Addressing these challenges requires further exploration and innovation, thereby enhancing the effectiveness of portfolio optimization.

There have been numerous research findings in this field, with a significant portion adopting heuristic strategies [6–8]. In 2021, Cura [9] proposed a heuristic method based on the artificial bee colony (ABC) algorithm to solve MV model for portfolio optimization problem with constraint constraints. The ABC algorithm with low population size (ABC-LP), demonstrated superior performance in portfolio combinations consisting of multiple indices. After that, Deng et al. [10] proposed an improved ABC algorithm for optimizing blockchain portfolios, ensuring a balance between investment returns and risks. In 2022, Li et al. [11] established an uncertain income portfolio optimization model from the perspective of expected utility maximization and designed an improved grey wolf optimizer (GWO) algorithm.

In recent years, quantum computing has been one of the fastest-growing and most promising fields. Compared to classical computing, quantum computing has theoretical advantages in areas such as encryption [12], industrial manufacturing [13], chemical [14]. For combinatorial optimization problems, quantum approximate optimization algorithm (QAOA) [15] and its variants [16] are specifically used to solve them. Hadfield et al. [17] proposed Quantum Alternating Operator Ansatz (QAOAz) to extend the QAOA framework to allow alternation between more general families of operators. Wang et al. [18] thoroughly discussed the application of QAOA in solving combinatorial optimization problems, particularly the maximum cut problem, and simplified the parameter optimization process using the Fermionic representation. Acampora et al. [19] applied a genetic algorithm to optimize the QAOA parameters for solving the MaxCut problem. Behera et al. [20] studied a wide range of vehicle traffic management problems and proposed algorithms based on QAOA to solve the vehicle routing problem. Pelofske et al. [21] compared QAOA with 1 and 2 layers and Quantum Annealing (QA) on 127-qubit problems, using D-Wave's quantum annealers, and finds that QA outperforms QAOA on the evaluated devices.

Nowadays, quantum computing brings innovation to various fields of financial services and provides decision-makers with more effective decision support [22–24]. In 2018, Rebentrost et al. [25] proposed a quantum algorithm for pricing financial derivatives using the Monte Carlo method and studied how to apply the amplitude estimation algorithm to achieve a high-confidence price estimate with a quadratic quantum acceleration in the number of steps required. In 2021, Daniel

et al. [26] proposed a quantum algorithm for estimating economic capital requirements with quadratic acceleration, analyzing the scaling of real-world problem sizes and expected running times under reasonable assumptions for future quantum computers. In 2022, Liu et al. [27] developed a novel market prediction method by integrating Elman neural networks with quantum mechanics and employed a double-chain quantum genetic algorithm to adjust the learning rate, enhancing the model's responsiveness to dynamic information, the model's effectiveness was confirmed through predictions of the closing prices in six stock markets. More recently, Canabarro et al. [28] utilized QAOA to solve the MV model for portfolio optimization problem, conducting experiments on stocks listed on the Brazilian Stock Exchange to obtain the target portfolio weights. N. Slate et al. [29] proposed a QAOAz for addressing the MV problem. QAOAz [17] improves upon the mixing operator of QAOA, enabling it to adapt to more complex search spaces. Although its performance can be limited due to potential biases introduced by the mixing operator over non-trivial feasible solution spaces, QAOAz ensures that constraints are always satisfied, with a zero probability of generating infeasible solutions. QAOAz shows performance advantages over classical algorithms in dealing with large search spaces, it leverages the principles of quantum computing, particularly the parallelism on qubits, allowing it to handle more complex optimization problems than classical algorithms. In practical applications requiring dynamic asset allocation over time, the optimization algorithm can be re-executed periodically (e.g. daily or weekly) using updated market data (covariance matrices and return forecasts) to generate dynamic adjustment strategies.

In the field of investment portfolios, the majority of researchers typically choose MV model for modeling, with little research on its improved models. However, MV model do not comprehensively consider the risk contribution of individual assets to the overall risk, which may lead to excessive exposure to the risk of a specific asset and fail to achieve effective risk diversification. The RP model addresses this issue well, but its implementation is more complex. It requires investors to accurately measure and manage the risk contribution of each asset. For this purpose, we are conducting research on the RP model, making its complex quadratic and quartic term calculations simpler through four-qubit and two-qubit entanglement operations, which significantly improves the optimization efficiency of the RP model.

We propose a quantum alternating operator ansatz for risk parity model (QAOAz-RP). Additionally, the optimization effect of QAOAz is closely related to the performance of classical optimizers. To enhance the performance of QAOAz, a Particle Swarm Optimization (PSO) optimizer is introduced to tune the parameters of the quantum circuits. Finally, we conduct a comparative study of problems based on the RP model and the MV model. The main contributions of this paper include four aspects:

- (1) For the more complex but comprehensive RP model compared to MV model, the problem definition of RP is given, and the corresponding QAOAz solution (QAOAz-RP) with detailed algorithm process is proposed for the first time.
- (2) The problem Hamiltonian H_p with the ZZZZ term is derived, and the corresponding quantum circuit $U(H_p, \gamma_i)$ is ingeniously constructed using the parity check method. Meanwhile, the quantum circuit $U(H_b, \beta_i)$ is constructed using the ring XY-mixer.

- (3) To improve the optimization performance of QAOAz, a PSO optimizer is utilized to tune the parameters of the quantum circuit, thereby improving the quality of the approximate solutions on both MV and RP models.
- (4) Experiment on RP model shows that the proposed QAOAz-RP method outperforms state-of-the-art algorithms in terms of optimization performance. Additionally, through experiments on the inclusion of the PSO optimizer, PSO optimizer is effective on both MV and RP models.

The reminder of this paper is organized as follows. First, a brief introduction of RP model and QAOAz is given in Section 2. Then, the model definition and formulation process of the proposed algorithms are illustrated in Section 3. Afterwards, in Section 4, we conduct experiments on the algorithms and discuss its effectiveness. Finally, Section 5 summarizes this paper and discusses the future research direction.

2. Preliminary

2.1. Risk parity

Asset allocation is a very important process in investment, and the classic method of asset allocation is Markowitz's MV model [2] which is typically used as a portfolio model to solve portfolio problem. The portfolio problem based on the MV model can be represented as:

$$\begin{aligned} \min_{w_i} \quad & \sigma_p^2 \\ \text{s.t.} \quad & R_p = \sum_{i=1}^n w_i \mathbb{E}(R_i) \\ & \sum_{i=1}^n w_i = 1 \end{aligned} \quad (1)$$

where σ_p^2 is the portfolio variance, w_i represents the investment weight, and $\mathbb{E}(R_i)$ represents the expected return.

This class of traditional methods focus on the overall risk of the investment portfolio and the optimization goal is to reduce the risk of the portfolio, but they do not fully consider the contribution of each asset to the overall risk, which may lead to excessive exposure to a particular asset and fail to achieve effective risk diversification.

In 2005, Qian [3] proposed the RP model and constructed the corresponding mathematical model. The RP model is an investment model based on the principle of risk equilibrium, which aims to diversify risk among multiple assets through risk allocation, thereby improving the risk-return ratio of the investment portfolio. Compared to the traditional MV model, the RP model does not rely on the prediction of asset return but uses historical volatility data to measure risk, making the allocation results more robust.

An investor's portfolio can be seen as a proportionate allocation of investments across multiple assets, which form the investor's alternative investment set. Suppose this set includes n different assets, and the investment weight of each asset is w_i , and $\Sigma = (\sigma_{ij}) \in R^{n \times n}$ is the covariance matrix of the investment portfolio. This vector form is convenient for analyzing and quantifying the investor's asset allocation strategy. According to the RP model [3], the contribution of each asset to the total risk can be expressed as:

$$TRC_i = w_i \frac{\text{cov}(r_i, r_p)}{\sigma_p} \quad (2)$$

where $\text{cov}(r_i, r_p)$ is the covariance matrix of asset i with the portfolio return. The risk-parity model requires selecting an appropriate capital allocation ratio w_i , such that the risk contribution of each asset in the portfolio is equal, i.e., $TRC_i = TRC_j$.

2.2. Quantum alternating operator ansatz

Fahri [30] proposed quantum computing based on adiabatic evolution, where the system remains in the ground state throughout the process, and the final state corresponds to the ground state of the target

Hamiltonian H_p , providing an effective approach to solving complex problems. Fig. 1 illustrates the evolution.

The system evolves from the ground state $|\phi_0\rangle$ of the initial state H_b to the ground state $|\phi_1\rangle$ of the final state H_p step by step. To explore this evolution, it can be represented mathematically as follows:

$$H(t) = (1 - \frac{t}{T})H_b + \frac{t}{T}H_p \quad (3)$$

where t represents the time elapsed during the evolution, T is the running time, $\tau = \frac{t}{T} \in [0, 1]$, and QAOA [15] follows the same principle based on the quantum adiabatic algorithm.

Quantum Alternating Operator Ansatz (QAOAz) [17] is a variant of the Quantum Approximate Optimization Algorithm. For constrained optimization problems, QAOA handles constraints by adding penalty terms to the objective function, whereas QAOAz incorporates the constraints into the mixer operator $U(H_b, \beta_i)$. Traditional X-mixers can cause quantum states to randomly walk through all possible states, which may not satisfy constraints like the total number of invested assets equaling B , thus easily moving out of the feasible solution space. In contrast, XY-mixers perform pairwise qubit XY rotations, confining the quantum walk to subspaces with a constant Hamming weight (the number of 1s in the bit string). This inherent property ensures that the number of selected assets in the portfolio remains constant.

The $|\phi_0\rangle$ in QAOAz should be a feasible state and the mixer operator $U(H_b, \beta_i)$ in QAOAz should ensure that the quantum walk occurs only within the space of feasible solutions, with zero probability for infeasible solutions. The overall process can be represented as:

$$|\phi_1\rangle = |\vec{\beta}, \vec{\gamma}\rangle = U(\vec{\beta}, \vec{\gamma})|\phi_0\rangle \quad (4)$$

where $|\phi_0\rangle$ is the initial state, $U(\vec{\beta}, \vec{\gamma})$ contains several depths of the quantum circuit, and $|\phi_1\rangle$ is the final state obtained after the quantum circuit.

In the QAOAz, the use of the XY mixer as the mixing operator H_b is a common and effective choice. As shown in Fig. 2, the ring and complete graph XY-mixers are two distinct methods for constructing mixing layers in quantum circuits. In a complete graph XY-mixer, every qubit interacts with all other qubits, resulting in a fully connected structure where all qubits are entangled during the mixing process. In contrast, the ring XY-mixer connects each qubit only to its two neighboring qubits, forming a cyclic structure.

Due to the reduced number of interactions, the ring XY-mixer yields a quantum circuit with lower depth compared to the complete graph XY-mixer, making it more efficient in terms of resource utilization and potentially less susceptible to noise in large-scale quantum circuits. For implementation on an all-to-all qubit device, the ring mixer can be implemented in depth $O(\log n)$ with no error [31], while the complete graph mixer can be implemented in query complexity $O(n^2/\sqrt{\delta})$ with error δ .

Overall, QAOAz is an application of quantum computing in the field of optimization. It combines the parallelism of quantum computing with the flexibility of classical optimization algorithms, aiming to leverage the unique properties of quantum computers to solve complex problems that are difficult for classical algorithms to handle. For clarity, the main symbols involved in the problem formulation, Hamiltonian construction, and quantum circuit design are summarized in Table 1.

3. PSO-QAOAz for portfolio optimization problem

In this section, we firstly define the RP model for the portfolio optimization problem, and then present the algorithm process of QAOAz for solving this problem, along with the derivation of problem Hamiltonian, and the construction of its quantum circuit. Finally, in order to improve the optimization performance of QAOAz, a PSO optimizer is utilized to tune the parameters of the quantum circuit, which is effective for both RP and MV models. Our algorithms were designed to automatically adjust portfolio weights based on preset periods (daily, weekly, or monthly), enabling real-time portfolio optimization and ensuring optimal asset allocation.

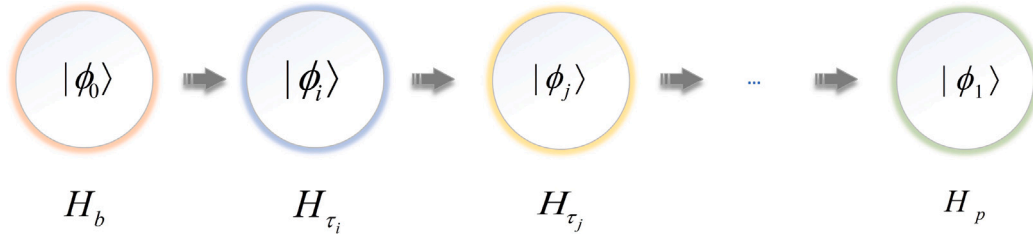


Fig. 1. Schematic diagram of the adiabatic quantum evolution process.

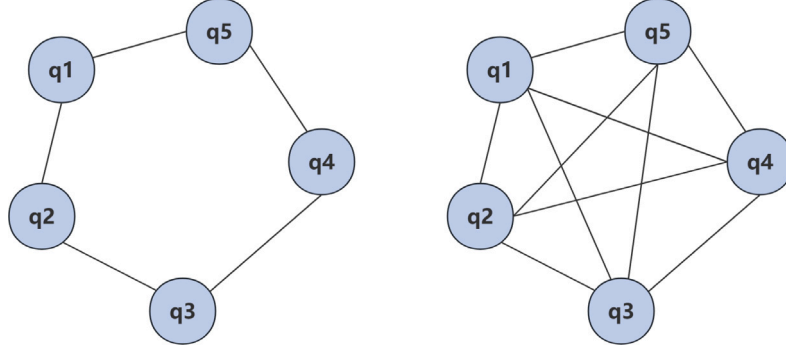


Fig. 2. The structure of the ring (left) and complete graph (right) XY-mixers, where each node represents a qubit and each edge represents the corresponding quantum gate.

Table 1

Definition of symbols.

Symbol	Description
w_i	Investment weight (continuous form)
x_i	Investment weight (binary form; selected asset indicator)
$\Sigma = (\sigma_{ij})$	Covariance matrix
μ_i	Expected return of the i th stock
TRC_i	Risk contribution of the i th stock
Z_i	Pauli-Z operator; Ising representation of binary variable
H_p	Problem Hamiltonian
H_b	Mixer Hamiltonian
β_i, γ_i	Mixing and phase parameters at layer i of QAOAz
A	Penalty coefficient for constraint enforcement
B	Constraint constant (number of selected assets)
p	Depth of the quantum circuit (QAOAz layers)

3.1. Problem definition

Based on Eq. (2) in Section 2.1, we can express the investment portfolio problem using the RP model [3] as follows:

$$\begin{aligned}
 & \min \sum_{i=1}^n \sum_{j=1}^n (TRC_i - TRC_j)^2 \\
 & = \sum_{i=1}^n \sum_{j=1}^n (w_i(\Sigma W)_i - w_j(\Sigma W)_j)^2 \\
 & \text{s.t. } \mathbf{1}^T W = 1
 \end{aligned} \quad (5)$$

where n represents the total amount of assets, $\Sigma \in R^{n \times n}$ is the covariance matrix of the investment portfolio, and $W = (w_1, w_2, \dots, w_n)$, w_i represents the investment weight of each asset.

In the binary representation of portfolio weights, whether a stock is included in the portfolio is determined by the value of the variable x_i , where x_i equals 1 indicates that the stock is included in the portfolio, and x_i equals 0 means that the stock is not included. When considering the discrete nature of these constraints, we can scale up the values of x_i proportionally to ensure they fall within the integer range. According to Eq. (5), the equivalent form of the portfolio optimization problem

can be stated as follows:

$$\begin{aligned}
 & \min \sum_{i=1}^n \sum_{j=1}^n (TRC_i - TRC_j)^2 \\
 & = \sum_{i=1}^n \sum_{j=1}^n (x_i(\Sigma X)_i - x_j(\Sigma X)_j)^2 \\
 & \text{s.t. } \mathbf{1}^T X = B
 \end{aligned} \quad (6)$$

where $X = (x_1, x_2, \dots, x_n)$, B represents the number of investment assets.

3.2. QAOAz for portfolio optimization problem

3.2.1. Algorithm process

As shown in Eq. (4), $U(\vec{\beta}, \vec{\gamma})$ is the quantum circuit part in QAOA, and it can be further divided into the unitary transformation with generator H_b and the unitary transformation with generator H_p , as follows:

$$U(\vec{\beta}, \vec{\gamma}) = \prod_{i=1}^m U(H_b, \beta_i) U(H_p, \gamma_i) \quad (7)$$

where m represents the number of evolutionary steps, and each step of the quantum circuit can be represented by the product of these two unitary transformations.

Subsequent parameter optimization involves optimizing the parameters within these two unitary transformations, with these parameters present at every depth of the quantum circuit. The subscript i of the two parameters indicates two parameters at a certain depth i of the quantum circuit. The specific expressions for the two unitary transformations are as follows:

$$U(H_b, \beta_i) = e^{-iH_b\beta_i} \quad (8)$$

$$U(H_p, \gamma_i) = e^{-iH_p\gamma_i} \quad (9)$$

where H_b is the mixer Hamiltonian and H_p is the problem Hamiltonian. The complete algorithm process is shown in Algorithm 1.

Therefore, we need to determine the Hamiltonian problem H_p based on x_i , which represents the binary decision variables for the portfolio weights. As shown in Eq. (10), we map it to the Ising Hamiltonian

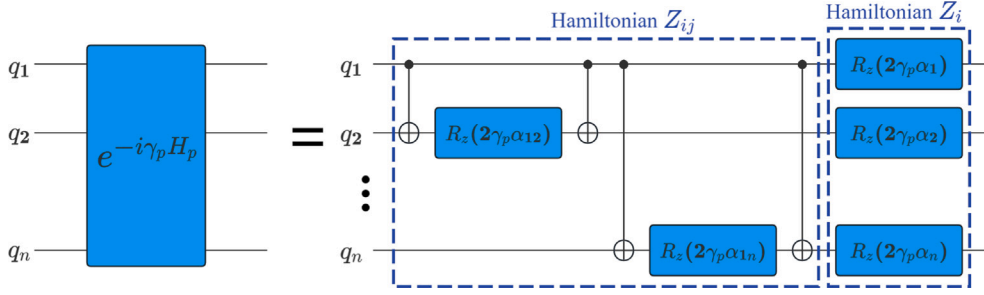


Fig. 3. Quantum circuit of the operator $e^{-i\gamma_p H_p}$ for a Hamiltonian with Z_i and $Z_i Z_j$ terms, implemented using a combination of R_z and CNOT gates.

Algorithm 1 QAOAz for RP model

Input: Problem Hamiltonian H_p of RP model

Output: Quantum state $|\varphi\rangle$

- 1: Prepare a feasible solution $|\varphi_0\rangle$, and a mixer Hamiltonian H_b
- 2: $H_b \leftarrow \sum_{i=1}^n X_i X_{i+1} + Y_i Y_{i+1}$
- 3: Initialize parameters $\gamma = \{\gamma_1, \gamma_2, \dots, \gamma_n\}$ and $\beta = \{\beta_1, \beta_2, \dots, \beta_n\}$
- 4: Construct the Quantum Circuit (QC) by alternately applying $U(H_b, \beta_i)$ and $U(H_p, \gamma_i)$ for n iterations.
- 5: **while** $\langle \varphi | H | \varphi \rangle$ essentially stops changing **do**
- 6: Obtain $|\varphi\rangle$ through QC
- 7: Calculate the expected value $\langle \varphi | H | \varphi \rangle$
- 8: Optimize γ and β in QC
- 9: **end while**
- 10: Measure QC with optimized parameters
- 11: Obtain the final quantum state $|\varphi\rangle$

form using the Pauli Z operators. Assuming the objective function is F , the obtained quantum operator is $\hat{F} = F(\hat{x}_1, \dots, \hat{x}_n)$. Obviously, the mapped problem Hamiltonian and the original objective function have the following relationship:

$$H_p |x_1, \dots, x_n\rangle = F(x_1, \dots, x_n) |x_1, \dots, x_n\rangle \quad (10)$$

Next, we will continue to derive the Hamiltonian corresponding to the combinatorial optimization model and the corresponding quantum circuit.

3.2.2. Derivation of problem hamiltonian

The translation of the RP model for portfolio optimization problem into the objective function is as follows:

$$F_{RP} = \sum_{i=1}^n \sum_{j=1}^n (x_i (\sum_{k=1}^n X_k) - x_j (\sum_{k=1}^n X_k))^2 + A (\sum_{i=1}^n x_i - B)^2 \quad (11)$$

where $A(\sum_{i=1}^n x_i - B)^2$ is a constraint term, A is the penalty coefficient. The constraint term is removed from the objective function and incorporated into the mixer. Simplifying Eq. (11) and removing the constraint terms gives the following result.

$$\begin{aligned} \hat{F}_{RP} &= \sum_{i=1}^n \sum_{j=1}^n (x_i (\sum_{k=1}^n X_k) - x_j (\sum_{k=1}^n X_k))^2 \\ &= \sum_{i=1}^n \sum_{j=1}^n (x_i \sum_{k=1}^n \sigma_{ik} x_k - x_j \sum_{k=1}^n \sigma_{jk} x_k)^2 \\ &= \sum_{i=1}^n \sum_{j=1}^n (x_i^2 \sum_{k,k'=1}^n \sigma_{ik} x_k \sigma_{ik'} x_{k'} - 2x_i x_j \sum_{k=1}^n \sum_{l=1}^n \sigma_{ik} \sigma_{jl} x_k x_l \\ &\quad + x_j^2 \sum_{l,l'=1}^n \sigma_{jl} x_l \sigma_{jl'} x_{l'}) \\ &= 2n \sum_{i=1}^n \sum_{k=1}^n \sum_{k'=1}^n \sigma_{ik} \sigma_{ik'} x_i x_k x_{k'} - 2 \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n \sigma_{ik} \sigma_{jl} x_i x_j x_k x_l \end{aligned} \quad (12)$$

When mapped to the problem Hamiltonian H_p by $x_i = \frac{1+Z_i}{2}$, it can be expressed as:

$$\begin{aligned} H_p &= \sum_{i=1}^n Z_i \left(\sum_{j=1}^n \sum_{k=1}^n \frac{n}{4} \sigma_{ij} \sigma_{ik} - \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n \frac{1}{4} \sigma_{ik} \sigma_{jl} \right) \\ &\quad + \sum_{j=1}^n Z_j \left(\sum_{i=1}^n \sum_{k=1}^n \frac{n}{2} \sigma_{ij} \sigma_{ik} - \sum_{i=1}^n \sum_{k=1}^n \sum_{l=1}^n \frac{1}{4} \sigma_{ik} \sigma_{jl} \right) \\ &\quad + \sum_{i=1}^n \sum_{j=1}^n Z_i Z_j \left(\sum_{k=1}^n \frac{n}{2} \sigma_{ij} \sigma_{ik} - \sum_{k=1}^n \sum_{l=1}^n \frac{1}{2} \sigma_{ik} \sigma_{jl} \right) \\ &\quad + \sum_{j=1}^n \sum_{k=1}^n Z_j Z_k \left(\sum_{i=1}^n \frac{n}{4} \sigma_{ij} \sigma_{ik} \right) + \sum_{i=1}^n \sum_{k=1}^n Z_i Z_k \left(- \sum_{j=1}^n \sum_{l=1}^n \frac{1}{8} \sigma_{ik} \sigma_{jl} \right) \\ &\quad + \sum_{j=1}^n \sum_{l=1}^n Z_j Z_l \left(- \sum_{i=1}^n \sum_{k=1}^n \frac{1}{8} \sigma_{ik} \sigma_{jl} \right) + \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n Z_i Z_j Z_k \left(\frac{n}{4} \sigma_{ij} \sigma_{ik} \right) \\ &\quad + \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n Z_i Z_j Z_k \left(- \sum_{l=1}^n \frac{1}{4} \sigma_{ik} \sigma_{jl} \right) + \sum_{i=1}^n \sum_{j=1}^n \sum_{l=1}^n Z_i Z_j Z_l \left(- \sum_{k=1}^n \frac{1}{4} \sigma_{ik} \sigma_{jl} \right) \end{aligned} \quad (13)$$

The covariance matrix is a symmetric matrix, which allows for some simplified calculations in the derivation. The complete derivation with the inclusion of the constraint terms can be found in [Appendix A](#).

3.2.3. Quantum circuit

Based on Eq. (13) in Section 3.2.2, the objective function of the RP model is assigned to quantum operators. As shown in Eq. (8) and (9), the quantum circuits $U(H_p, \gamma_i)$ and $U(H_b, \beta_i)$, corresponding to the problem Hamiltonian H_p and the mixed Hamiltonian H_b , respectively, are constructed as follows.

(1) Quantum circuit $U(H_p, \gamma_i)$

Quantum circuit for simulating the Hamiltonian of Z_i and $Z_i Z_j$ as shown in Fig. 3. We implemented the Hamiltonians Z_i and Z_{ij} using a combination of the rotation operators about the $\hat{z}$ axes (R_z) and CNOT gates. For the Hamiltonian Z_i , we apply the R_z on the i th qubit. For the Hamiltonian Z_{ij} , we apply a CNOT gate with the i th qubit as the control and the j th qubit as the target. Then, a R_z is applied to the j th qubit, followed by a repetition of the previous CNOT operations.

In addressing the Hamiltonian with quartic terms, existing approaches typically recode them as $Z_{ij} = Z_i Z_j$ terms and incorporate additional constraints [16]. This process breaks down higher-order terms into quadratic ones with auxiliary variables and penalty terms, thereby ensuring equivalence. However, this method demands extra qubits and quantum gates. Conversely, directly implementing $Z_i Z_j Z_k Z_l$ terms not only reduces qubit requirements but also maintains the inherent complex interaction dependencies of the problem.

Inspired by the parity check method [32], quantum circuit for simulating the Hamiltonian of $Z_i Z_j Z_k$ and $Z_i Z_j Z_k Z_l$ is illustrated by the example of $Z_1 Z_2 Z_3$ and $Z_1 Z_2 Z_3 Z_4$. As shown in Fig. 4, $|0\rangle$ refers to the auxiliary qubit, and α represents the coefficient corresponding to each quantum gate in Section 3.2.2.

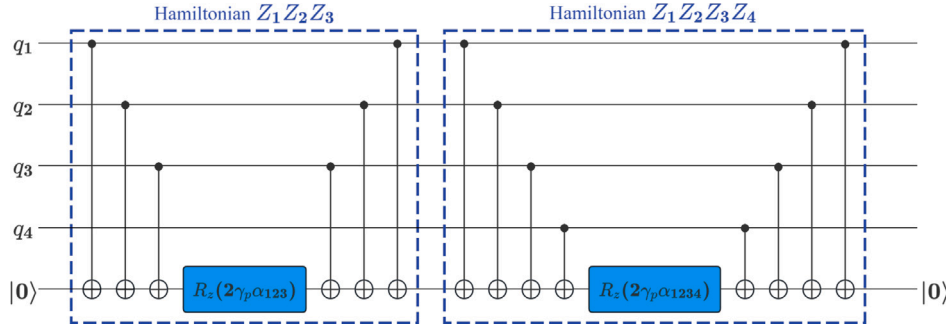


Fig. 4. Quantum circuit for simulating the Hamiltonian $Z_1Z_2Z_3$ and $Z_1Z_2Z_3Z_4$, where the blue boxes on the left and right correspond to the specific implementations of $Z_1Z_2Z_3$ and $Z_1Z_2Z_3Z_4$, respectively.

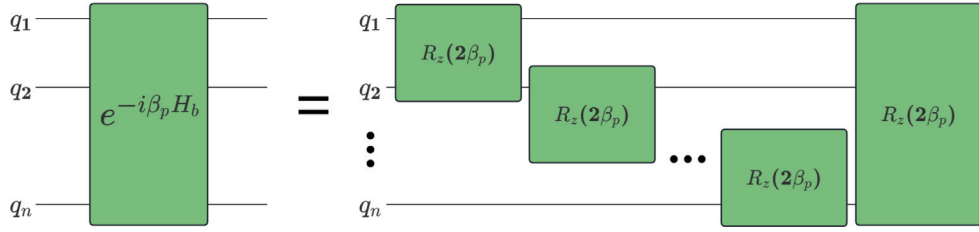


Fig. 5. Quantum circuit of the operator $U(H_b, \beta_i)$, where $U(H_b, \beta_p) = e^{-i\beta_p H_b}$.

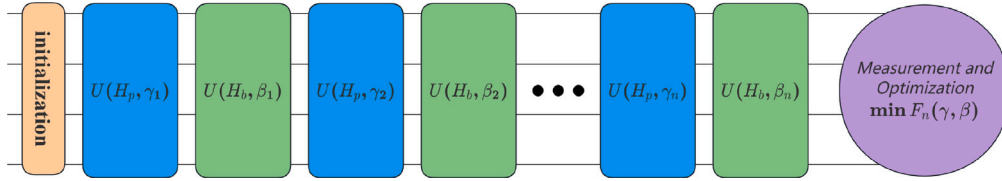


Fig. 6. Quantum circuit of QAOAz.

For the Hamiltonian $Z_1Z_2Z_3$, CNOT gates are applied with the 1st, 2nd, and 3rd qubits as control qubits and the auxiliary qubit as the target. Then, a R_z is applied to the auxiliary qubit, followed by a repetition of the previous CNOT operations. The same procedure is applied for the Hamiltonian $Z_1Z_2Z_3Z_4$, with the addition of an extra CNOT gate pair, where the 4th qubit serves as the control and the auxiliary qubit as the target.

(2) Quantum circuit $U(H_b, \beta_i)$

As shown in Fig. 5, we replaced the traditional X-Mixer with a ring XY-Mixer, which preserves the Hamming weight of quantum states. This design is particularly well-suited for solving constrained optimization problems.

The quantum circuit part of QAOAz is composed of the product of two unitary transformations, which correspond to the quantum gates of the Hamiltonian H_p and H_b , respectively. The complete circuit is shown in Fig. 6, which alternates between the phase separation operator $U(H_p, \gamma_i)$ and the mixing operator $U(H_b, \beta_i)$ after initialization.

3.3. PSO optimizer for QAOAz

For both QAOAz-RP and QAOAz-MV, we used the PSO optimizer to optimize the parameters of the quantum circuits. PSO algorithm is a random search algorithm based on population collaboration. The derivative-free optimizers are generally more effective in terms of given computational resources, problem size, and circuit depth. When a certain number of circuit parameters are involved, meta-heuristic methods can more efficiently search for the required circuit parameters compared to other gradient-free optimizers. During the parameter

optimization process, we design PSO as a classical optimizer. With the increase in the number of circuit parameters, meta-heuristic algorithms can often provide more effective search and solutions for these multi-parameter problems. The complete algorithm process is shown in Algorithm 2.

Algorithm 2 PSO optimizer for QAOAz

Input: Circuit parameters of QAOAz: $[\gamma_1, \dots, \gamma_n, \beta_1, \dots, \beta_n]$

Output: $gBest$

```

1: for each particle  $i$  (circuit parameters) do
2:   Initialize velocity  $v_i$  and position  $x_i$  for particle  $i$ 
3:   Evaluate particle  $i$ , and set  $pBest_i = x_i$ 
4: end for
5:  $gBest = \min\{pBest_i\}$ 
6: while not stop do
7:   for  $i=1$  to  $N$  do
8:     Update the velocity and position of particle  $i$ 
9:     Quantum circuit measurement and objective function calculation
10:    if  $\text{fit}(x_i) < \text{fit}(pBest_i)$  then
11:       $pBest_i = x_i$ ;
12:    end if
13:    if  $\text{fit}(pBest_i) < \text{fit}(x_i)$  then
14:       $gBest = pBest_i$ ;
15:    end if
16:  end for
17: end while

```

Table 2
Selected investment portfolio samples.

Investment portfolio	Stock codes					
Portfolio 1	600,926	000,786	002,001	000,877	600,741	300,979
Portfolio 2	600,547	000,963	002,007	000,800	600,585	300,750
Portfolio 3	600,938	000,977	002,371	000,895	600,941	300,751
Portfolio 4	600,104	000,776	002,074	000,983	600,061	300,347
Portfolio 5	600,188	000,538	002,230	000,708	600,346	300,661
Portfolio 6	600,000	000,999	002,500	000,001	600,023	300,001
Portfolio 7	AAPL	MSFT	GOOGL	AMZN	NVDA	META
Portfolio 8	TSLA	JPM	BRK-B	V	JNJ	UNH
Portfolio 9	AIR.PA	OR.PA	SAN.PA	DTE.DE	BAS.DE	BMW.DE
Portfolio 10	VOW3.DE	SIE.DE	ADS.DE	AZN.L	HSBA.L	RIO.L
Portfolio 11	EEM	EWZ	FXI	INDA	VWO	RSX
Portfolio 12	TCEHY	BABA	JD	INFY	PDD	YPF

The final $gBest$ represents the optimal parameter obtained after PSO optimizer. The [update of the velocity and position in the d th dimension of particle i is a key step in the evolutionary process of the algorithm. This process can be described by the following mathematical expression:

$$\begin{cases} v_i^{d+1} = w \times v_i^d + c_1 \times \text{rand}_1^d \times (pBest_i^d - x_i^d) + c_2 \times \text{rand}_2^d \times (gBest^d - l_i^d) \\ l_i^{d+1} = l_i^d + v_i^{d+1} \end{cases} \quad (14)$$

where v_i is the velocity vector, l_i is the position vector, i is the particle number, and d is the dimension of the information carried by the individual particle.

When using PSO to optimize the QAOAz quantum circuit, the goal is to minimize the loss function. The computation of the fitness function requires the participation of a quantum device. Firstly, the quantum circuit generates the binary investment weights corresponding to each asset, and then the function value is calculated. The fitness function value is obtained by substituting the values into Eqs. (11) and (16) through the quantum circuit.

4. Experiment

4.1. Datasets

We conducted experiments on the RP model and the MV model using multiple portfolios. Current quantum computing hardware has limited capabilities, so to ensure the feasibility of the experiment, six specific stocks are selected for each portfolio. In Portfolios 1–6, we utilized data from Chinese A-shares, where each portfolio consists of stocks randomly selected from the CSI-300 Composite Index of the Chinese A-share market. Portfolios 7–8 are based on U.S. stock market data. Portfolios 9–10 consist of European market data, while Portfolios 11–12 incorporate emerging market data. The sample data is from the entire year of January 1, 2022 to December 31, 2022 (see Table 2). The covariance matrix of portfolios 1–6 is presented in Appendix B.

4.2. Experiment settings

The experimental environment is equipped with CentOS 8 operating system, an Intel Xeon Gold 5218 CPU, and a NVIDIA GeForce RTX 2080Ti graphics card. The code is written in Python, primarily utilizing the framework Cirq, and its source file is available on GitHub.¹

In the QAOAz algorithm, we have set the depth of the quantum circuit to 5, with parameter values ranging from 0 to 2π . The depth

Table 3
Parameter configurations for all algorithms.

Algorithm	Parameters	
GA [9]	$max_iterations = 100$	$population_size = 50$
	$crossover_rate = 0.9$	$mutation_rate = 0.05$
ABC-LP [11]	$max_iterations = 100$	$food_sources = 20$
	$employed_bees = 5$	$onlooker_bees = 5$
GWO [11]	$max_iterations = 300$	$population_size = 50$
QAOA [28]	$P=5$	$\lambda, \beta \in (0, 2\pi)$
QAOAz [29]	$P=5$	$\lambda, \beta \in (0, 2\pi)$
PSO-QAOAz	$P=5$	$\lambda, \beta \in (0, 2\pi)$
	$max_iterations = 10$	$num_particles = 30$
	$inertia = 0.7$	$cognitive_constant = 1.49$
	$social_constant = 1.49$	

configuration of the proposed QAOAz-RP aligns with that of the existing QAOA [28] and QAOAz [29] employed for solving the MV model. In the PSO algorithm, we iterate 10 times, employ 30 particles, set the inertia coefficient to 0.7, the individual learning factor to 1.49, and the group learning factor to 1.49. The complete parameters and comparison algorithm parameters are shown in Table 3.

4.3. Results

To verify the superiority of the proposed algorithm, we conducted comparative experiments on both RP model and MV model. We selected the swarm intelligence algorithms ABC-LP [9], GWO [11], and GA [11] as the benchmark algorithms, which have all shown good performance in the latest research on portfolio optimization. We ran each algorithm 10 times and recorded the best, worst, average results of the optimization and the standard deviation. The optimization objective function is defined as Eqs. (11) and (16).

In the experiment based on the RP model, we conducted experimental analyses of both QAOAz-RP and PSO-QAOAz-RP, and introduced the Approximate Ratio (AR), which provides a clearer indication of the effectiveness of each algorithm's optimization. The approximate ratio is defined as follows:

$$AR = \min\left\{\frac{c^*}{c}, \frac{c}{c^*}\right\} \quad (15)$$

where c^* is the objective function value obtained by repeating the approximate algorithm, and c is the optimal value of the optimization problem.

Additionally, in the experiment based on the MV model, we compared the proposed algorithm with existing QAOA [28] and QAOAz [29] to demonstrate the superiority of the PSO algorithm in optimizing QAOAz quantum circuits.

4.3.1. Experiment for portfolio optimization problem with RP model

As shown in Table 4 and Fig. 7, in the optimization experiments based on the RP model, the average optimization value of the QAOAz-RP on the selected portfolio samples is superior to that of the baseline model in many portfolios, along with the best corresponding AR. PSO-QAOAz-RP utilizes a PSO optimizer to tune the parameters in QAOAz-RP, making the algorithm more efficient and achieving better optimization results. In portfolio 7–8, our quantum method underperforms classical ones. This might be due to the significant differences among these market portfolios. The optimal RP expectations for these portfolios are much lower than others, QAOAz-RP and PSO-QAOAz-RP may therefore be prone to local optima. We also calculated the standard deviation of the performance for each algorithm. The standard deviation of the optimization value for PSO-QAOAz-RP is consistently lower than that of the other methods, demonstrating the stability of our algorithm. The approximation ratios, along with the median and overall range of optimization values, demonstrate that QAOAz-RP and PSO-QAOAz-RP outperform the baseline algorithm in most portfolios, highlighting our algorithms superior performance in solving optimization problems based on the RP model.

¹ <https://github.com/MaALPS/QAOA-Portfolio>

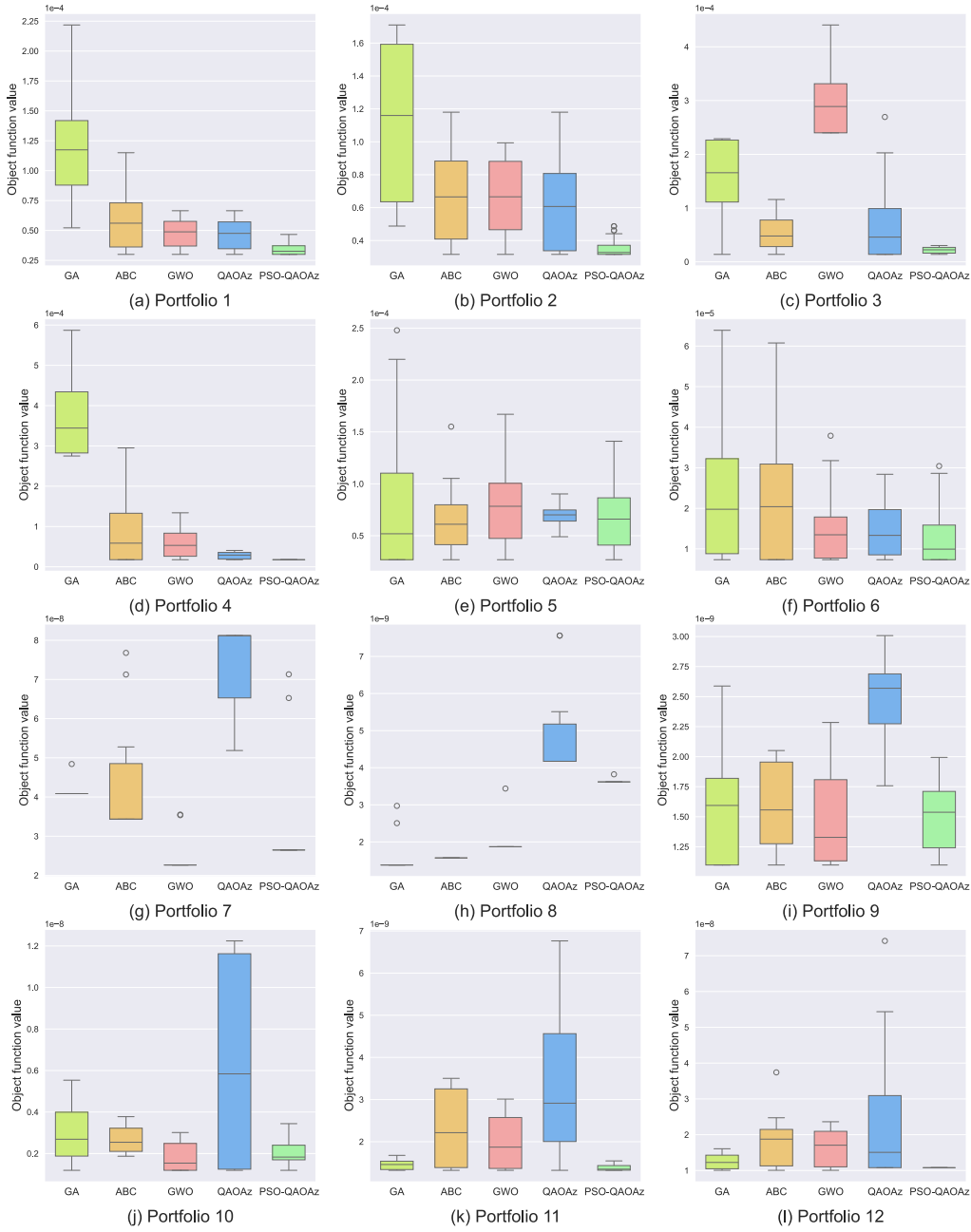


Fig. 7. Boxplot of F_{RP} values among different algorithms for RP model.

4.3.2. Experiment for portfolio optimization problem with MV model

We used the PSO optimizer on QAOAz to conduct experiments on the MV model, demonstrating the effectiveness of the PSO optimizer. The definition of the MV model and the derivation of the Hamiltonian can be found in [Appendix C](#), where the value of q is set to 0.5. It should be noted that the objective function is

$$F_{MV} = q \sum_{i,j=1}^n x_i x_j \sigma_{ij} - (1-q) \sum_{i=1}^n x_i \mu_i + A \left(\sum_{i=1}^n x_i - B \right)^2 \quad (16)$$

As shown in [Table 5](#) and [Fig. 8](#), in the optimization experiment based on the MV model, the PSO-QAOAz-MV demonstrated the best average optimization value performance in portfolio 1–6. PSO-QAOAz-MV consistently achieves better objective function values compared to QAOA [28] and QAOAz [29], demonstrating superior optimization results. Notably, the standard deviation is also significantly lower than that of other algorithms, which is attributed to the powerful parameter

optimization capability of PSO, and experiments have demonstrated that the PSO optimizer is able to improve the stability of the algorithm.

Overall, the experiment on RP model for portfolio optimization problem shows that the proposed QAOAz method outperforms state-of-the-art algorithms. Additionally, through experiments on the PSO optimizer, PSO optimizer is effective on both MV and RP models for portfolio optimization problem.

4.3.3. Hyperparameter discussion

The selection of iterations for the PSO algorithm aims to balance optimization effectiveness with computational efficiency. As shown in [Fig. 9](#), systematic convergence testing on Portfolio 1 showed that 10 iterations were sufficient to improve algorithmic performance without incurring excessive computational time, and to achieve a stable optimization result within a reasonable time.

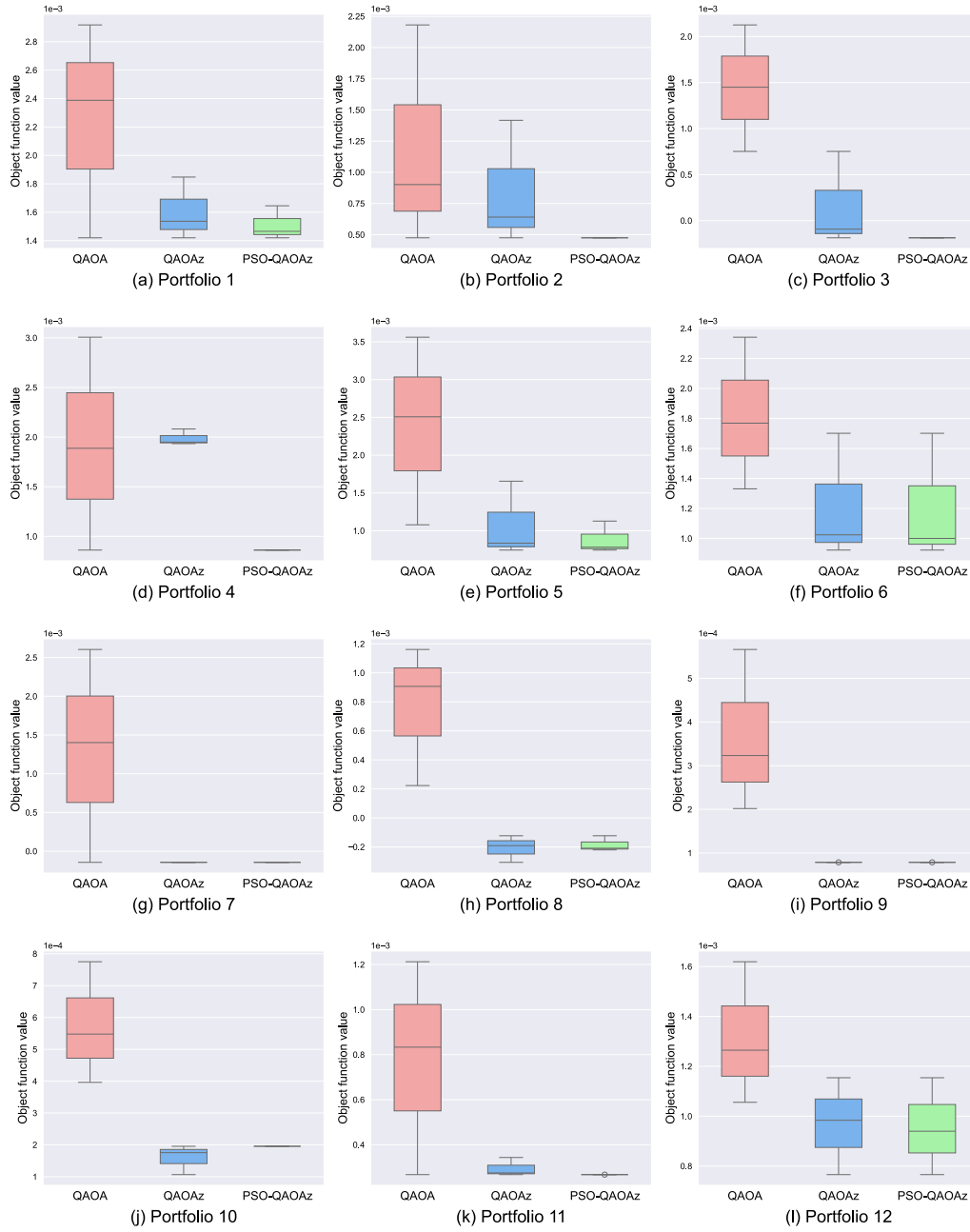
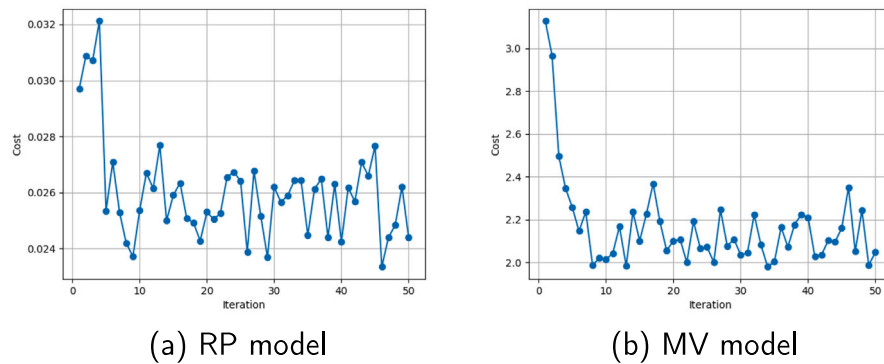
Fig. 8. Boxplot of F_{MV} among different algorithms for MV model.

Fig. 9. Convergence evaluation of PSO Iterations.

Table 4

Experiment results of our QAOAz-RP and PSO-QAOAz-RP algorithms for the RP model compared with GA, ABC-LP and GWO. The best results are in bold and the second-best results are underlined.

Portfolio	Methods	Objective function F_{RP}				AR
		Optimal	Worst	Average	Standard deviation	
Portfolio 1	GA	$5.22 * 10^{-5}$	$2.43 * 10^{-4}$	$1.21 * 10^{-4}$	$4.71 * 10^{-5}$	24.81%
	ABC-LP	$3.00 * 10^{-5}$	$1.15 * 10^{-4}$	$5.30 * 10^{-5}$	$3.29 * 10^{-5}$	56.50%
	GWO	$3.00 * 10^{-5}$	$6.65 * 10^{-5}$	$4.62 * 10^{-5}$	$1.50 * 10^{-5}$	64.94%
	QAOAz-RP	$3.00 * 10^{-5}$	<u>$6.65 * 10^{-5}$</u>	<u>$4.45 * 10^{-5}$</u>	<u>$1.46 * 10^{-5}$</u>	67.57%
	PSO-QAOAz-RP	$3.00 * 10^{-5}$	$5.22 * 10^{-5}$	$3.23 * 10^{-5}$	$6.83 * 10^{-6}$	92.88%
Portfolio 2	GA	$4.88 * 10^{-5}$	$1.71 * 10^{-4}$	$1.10 * 10^{-4}$	$6.44 * 10^{-5}$	28.73%
	ABC-LP	$3.16 * 10^{-5}$	$1.18 * 10^{-4}$	$6.22 * 10^{-5}$	$3.53 * 10^{-5}$	50.76%
	GWO	$3.16 * 10^{-5}$	$9.94 * 10^{-5}$	$6.44 * 10^{-5}$	$2.66 * 10^{-5}$	49.02%
	QAOAz-RP	$3.16 * 10^{-5}$	<u>$1.18 * 10^{-4}$</u>	<u>$6.22 * 10^{-5}$</u>	<u>$3.53 * 10^{-5}$</u>	50.76%
	PSO-QAOAz-RP	$3.16 * 10^{-5}$	$4.87 * 10^{-5}$	$3.33 * 10^{-5}$	$5.44 * 10^{-6}$	94.89%
Portfolio 3	GA	$1.39 * 10^{-5}$	$2.29 * 10^{-4}$	$1.88 * 10^{-4}$	$8.70 * 10^{-5}$	7.39%
	ABC-LP	$1.39 * 10^{-5}$	<u>$1.16 * 10^{-4}$</u>	$5.20 * 10^{-5}$	<u>$3.64 * 10^{-5}$</u>	26.74%
	GWO	$2.40 * 10^{-4}$	$5.01 * 10^{-4}$	$2.91 * 10^{-4}$	$7.80 * 10^{-5}$	4.78%
	QAOAz-RP	$1.39 * 10^{-5}$	$2.90 * 10^{-4}$	<u>$5.17 * 10^{-5}$</u>	$8.40 * 10^{-5}$	<u>26.88%</u>
	PSO-QAOAz-RP	$1.39 * 10^{-5}$	$3.04 * 10^{-5}$	$2.22 * 10^{-5}$	$8.69 * 10^{-6}$	62.89%
Portfolio 4	GA	$2.75 * 10^{-4}$	$5.87 * 10^{-4}$	$3.55 * 10^{-4}$	$1.23 * 10^{-4}$	7.75%
	ABC-LP	$1.76 * 10^{-5}$	$2.95 * 10^{-4}$	$7.36 * 10^{-5}$	$9.44 * 10^{-5}$	23.92%
	GWO	$1.76 * 10^{-5}$	$1.34 * 10^{-4}$	$6.13 * 10^{-5}$	$4.56 * 10^{-5}$	28.74%
	QAOAz-RP	$1.76 * 10^{-5}$	<u>$4.06 * 10^{-5}$</u>	<u>$2.68 * 10^{-5}$</u>	<u>$1.19 * 10^{-5}$</u>	65.79%
	PSO-QAOAz-RP	$1.76 * 10^{-5}$	$1.76 * 10^{-5}$	$1.76 * 10^{-5}$	0.00	100.00%
Portfolio 5	GA	$2.69 * 10^{-5}$	$2.88 * 10^{-4}$	$6.96 * 10^{-5}$	$8.03 * 10^{-5}$	38.65%
	ABC-LP	$2.69 * 10^{-5}$	$9.26 * 10^{-4}$	<u>$6.27 * 10^{-5}$</u>	<u>$2.81 * 10^{-5}$</u>	42.92%
	GWO	$2.69 * 10^{-5}$	$1.67 * 10^{-4}$	$6.75 * 10^{-5}$	$4.06 * 10^{-5}$	39.84%
	QAOAz-RP	$2.69 * 10^{-5}$	$1.41 * 10^{-4}$	$7.14 * 10^{-5}$	$9.35 * 10^{-6}$	37.68%
	PSO-QAOAz-RP	$2.69 * 10^{-5}$	$1.41 * 10^{-4}$	$6.03 * 10^{-5}$	$3.47 * 10^{-5}$	44.64%
Portfolio 6	GA	$7.34 * 10^{-6}$	$6.52 * 10^{-5}$	$2.13 * 10^{-5}$	$1.77 * 10^{-5}$	34.46%
	ABC-LP	$7.34 * 10^{-6}$	$6.52 * 10^{-5}$	$1.85 * 10^{-5}$	$1.77 * 10^{-5}$	39.68%
	GWO	$7.34 * 10^{-6}$	$2.84 * 10^{-4}$	<u>$1.29 * 10^{-5}$</u>	$9.15 * 10^{-6}$	56.90%
	QAOAz-RP	$7.34 * 10^{-6}$	$2.84 * 10^{-4}$	$1.36 * 10^{-5}$	<u>$8.36 * 10^{-6}$</u>	53.97%
	PSO-QAOAz-RP	$7.34 * 10^{-6}$	$2.84 * 10^{-4}$	$1.08 * 10^{-5}$	$7.56 * 10^{-6}$	67.96%

Table 5

Experiment results of our PSO-QAOAz-MV algorithm for the MV model compared with QAOA and QAOAz.

Portfolio	Methods	Objective function F_{MV}				AR
		Optimal	Worst	Average	Standard deviation	
Portfolio 1	QAOA	$1.42 * 10^{-3}$	$2.92 * 10^{-3}$	$2.39 * 10^{-3}$	$6.13 * 10^{-4}$	59.41%
	QAOAz	$1.42 * 10^{-3}$	$1.85 * 10^{-3}$	$1.54 * 10^{-3}$	$1.45 * 10^{-4}$	92.21%
	PSO-QAOAz-MV	$1.42 * 10^{-3}$	$1.65 * 10^{-3}$	$1.47 * 10^{-3}$	$8.03 * 10^{-5}$	92.21%
Portfolio 2	QAOA	$4.75 * 10^{-4}$	$2.18 * 10^{-3}$	$9.02 * 10^{-4}$	$8.53 * 10^{-4}$	52.66%
	QAOAz	$4.75 * 10^{-4}$	$1.42 * 10^{-3}$	$6.41 * 10^{-4}$	$3.54 * 10^{-4}$	74.10%
	PSO-QAOAz-MV	$4.75 * 10^{-4}$	$4.75 * 10^{-4}$	$4.75 * 10^{-4}$	0.00	100%
Portfolio 3	QAOA	$7.52 * 10^{-4}$	$2.13 * 10^{-3}$	$1.45 * 10^{-3}$	$7.82 * 10^{-4}$	11.42%
	QAOAz	$-1.87 * 10^{-4}$	$7.52 * 10^{-4}$	$-9.27 * 10^{-5}$	$2.97 * 10^{-4}$	49.57%
	PSO-QAOAz-MV	$-1.87 * 10^{-4}$	$-1.87 * 10^{-4}$	$-1.87 * 10^{-4}$	0.00	100%
Portfolio 4	QAOA	$8.62 * 10^{-4}$	$3.01 * 10^{-3}$	$1.89 * 10^{-3}$	$1.05 * 10^{-3}$	45.61%
	QAOAz	$8.62 * 10^{-4}$	$1.63 * 10^{-3}$	$9.38 * 10^{-4}$	$2.42 * 10^{-4}$	91.90%
	PSO-QAOAz-MV	$8.62 * 10^{-4}$	$8.62 * 10^{-4}$	$8.62 * 10^{-4}$	0.00	100%
Portfolio 5	QAOA	$1.08 * 10^{-3}$	$3.56 * 10^{-3}$	$2.51 * 10^{-3}$	$1.05 * 10^{-3}$	29.64%
	QAOAz	$7.44 * 10^{-4}$	$1.65 * 10^{-3}$	$8.35 * 10^{-4}$	$2.28 * 10^{-4}$	89.10%
	PSO-QAOAz-MV	$7.44 * 10^{-4}$	$1.13 * 10^{-3}$	$7.82 * 10^{-4}$	$1.21 * 10^{-4}$	95.14%
Portfolio 6	QAOA	$1.33 * 10^{-3}$	$2.34 * 10^{-3}$	$1.77 * 10^{-3}$	$4.20 * 10^{-4}$	52.15%
	QAOAz	$9.23 * 10^{-4}$	$1.70 * 10^{-3}$	$1.02 * 10^{-3}$	$2.49 * 10^{-4}$	90.49%
	PSO-QAOAz-MV	$9.23 * 10^{-4}$	$1.70 * 10^{-3}$	$1.00 * 10^{-3}$	$2.46 * 10^{-4}$	92.30%

The QAOAz-RP circuit depth is set to 5. As P increases, the computational cost and runtime grow exponentially, and quantum noise accumulation intensifies, which poses a significant challenge for the feasibility of current quantum hardware [16]. As shown in Fig. 10(a), $P = 5$ offers a reasonable trade-off. It balances circuit complexity, solution quality, hardware fault tolerance, and execution time.

5. Conclusion

In this paper, we presented PSO-QAOAz-RP and PSO-QAOAz-MV or solving portfolio optimization problems, with a carefully derived

Hamiltonian and an innovative quantum circuit design. By incorporating a PSO optimizer, the method improves parameter tuning and optimization performance of QAOAz, specifically applied to both the MV and RP models. The proposed method provides valuable insights into how quantum optimization can be effectively applied to real-world financial decision-making.

While our quantum optimization-based portfolio research shows great potential, several limitations offer avenues for future improvement. First, this study does not account for quantum noise and error, which can disrupt the coherence of quantum systems and compromise computational accuracy. Second, the focus is on short-term portfolio

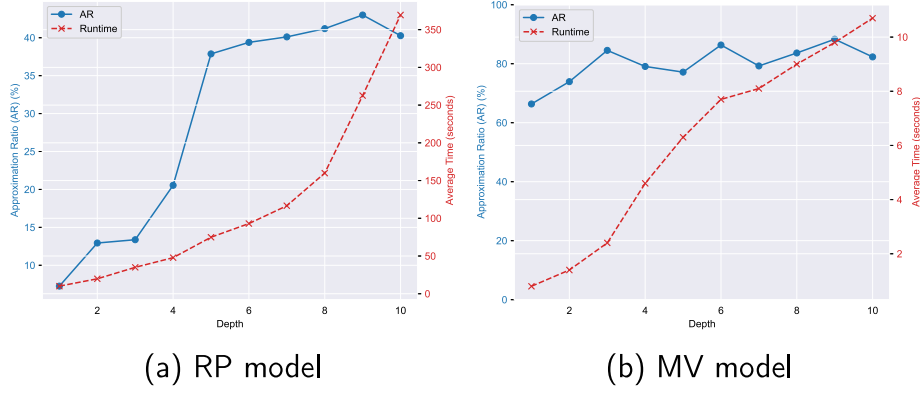


Fig. 10. Variation of approximation ratio and runtime with depth in QAOAz.

optimization, with no analysis of long-term market dynamics and the impact of external factors on portfolios. Future research could aim to extend the applicability of quantum algorithms to accommodate both short- and long-term investment objectives, thereby offering a more comprehensive reflection of market complexity. Finally, this study mainly concentrates on portfolio optimization, while the potential of quantum algorithms in finance is much broader. For future work, it is promising to explore how QAOA and other quantum algorithms can be applied to more financial areas, including asset pricing [33], risk assessment [34], and financial crisis prediction [35,36].

CRediT authorship contribution statement

Wenjie Liu: Writing – review & editing, Supervision, Conceptualization. **Yue Ma:** Writing – original draft, Software, Methodology, Data curation. **Yuchen Gu:** Writing – original draft, Formal analysis, Data curation. **Jiajun Cheng:** Validation, Software. **Qingshan Wu:** Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Complete derivation of problem Hamiltonian H_p

The complete derivation of the Hamiltonian H_p associated with the objective function in the RP model in Section 3, which also incorporates the constraint terms, can be detailed as follows.

$$\begin{aligned}
 & \sum_{i=1}^n \sum_{j=1}^n (x_i (\sum_{k=1}^n x_k) - x_j (\sum_{k=1}^n x_k))^2 + A (\sum_{i=1}^n x_i - B)^2 \\
 &= \sum_{i=1}^n \sum_{j=1}^n (x_i \sum_{k=1}^n \sigma_{ik} x_k - x_j \sum_{k=1}^n \sigma_{jk} x_k)^2 + A (\sum_{i=1}^n x_i - B)^2 \\
 &= \sum_{i=1}^n \sum_{j=1}^n (x_i^2 \sum_{k,k'=1}^n \sigma_{ik} \sigma_{ik'} x_{k'} x_{k'} - 2x_i x_j \sum_{k=1}^n \sigma_{ik} \sigma_{jl} x_k x_l + x_j^2 \sum_{l,l'=1}^n \sigma_{jl} \sigma_{jl'} x_l x_{l'}) \\
 &+ A (\sum_{i=1}^n x_i - B)^2 \\
 &= 2n \sum_{i=1}^n \sum_{k=1}^n \sum_{k'=1}^n \sigma_{ik} \sigma_{ik'} x_i^2 x_k x_{k'} - 2 \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n \sigma_{ik} \sigma_{jl} x_i x_j x_k x_l + A (\sum_{i=1}^n x_i - B)^2
 \end{aligned}$$

$$\begin{aligned}
 &= 2n \sum_{i=1}^n \sum_{k=1}^n \sum_{k'=1}^n \sigma_{ik} \sigma_{ik'} x_i x_k x_{k'} - 2 \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n \sigma_{ik} \sigma_{jl} x_i x_j x_k x_l + A (\sum_{i=1}^n x_i - B)^2 \\
 &= 2n \sum_{i=1}^n \sum_{k=1}^n \sum_{k'=1}^n \frac{1}{8} \sigma_{ik} \sigma_{ik'} (I + Z_i + Z_k + Z_i Z_k + Z_{k'} + Z_i Z_{k'} + Z_k Z_{k'} + Z_i Z_k Z_{k'}) \\
 &- 2 \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n \frac{1}{16} \sigma_{ik} \sigma_{jl} (I + Z_i + Z_j + Z_k + Z_l + Z_i Z_j + Z_i Z_k + Z_j Z_k + Z_i Z_l \\
 &+ Z_j Z_l + Z_k Z_l + Z_i Z_j Z_k + Z_i Z_j Z_l + Z_i Z_k Z_l + Z_j Z_k Z_l + Z_i Z_j Z_k Z_l) \\
 &+ A [\frac{1}{4} \sum_{i,j=1}^n (I + Z_i)(I + Z_j) - B \sum_{i=1}^n (I + Z_i)] \\
 &= \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{n}{4} \sigma_{ij} \sigma_{ik} (Z_i + Z_k + Z_j) + \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{n}{4} \sigma_{ij} \sigma_{ik} (Z_i Z_k + Z_i Z_j + Z_j Z_k) \\
 &+ \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{n}{4} \sigma_{ij} \sigma_{ik} (Z_i Z_j Z_k) \\
 &- \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n \frac{1}{8} \sigma_{ik} \sigma_{jl} (Z_i + Z_j + Z_k + Z_l + Z_i Z_j + Z_i Z_k + Z_j Z_k + Z_i Z_l + Z_j Z_l \\
 &+ Z_k Z_l + Z_i Z_j Z_k + Z_i Z_j Z_l + Z_i Z_k Z_l + Z_j Z_k Z_l + Z_i Z_j Z_k Z_l) \\
 &+ A [\frac{1}{4} \sum_{i,j=1}^n (I + Z_i)(I + Z_j) - B \sum_{i=1}^n (I + Z_i)] \\
 &= \sum_{i=1}^n Z_i \left(\sum_{j=1}^n \sum_{k=1}^n \frac{n}{4} \sigma_{ij} \sigma_{ik} \right) \\
 &+ 2 \sum_{j=1}^n Z_j \left(\sum_{i=1}^n \sum_{k=1}^n \frac{n}{4} \sigma_{ij} \sigma_{ik} \right) + 2 \sum_{i=1}^n \sum_{j=1}^n Z_i Z_j \left(\sum_{k=1}^n \frac{n}{4} \sigma_{ik} \sigma_{jk} \right) \\
 &+ \sum_{j=1}^n \sum_{k=1}^n Z_j Z_k \left(\sum_{i=1}^n \frac{n}{4} \sigma_{ij} \sigma_{ik} \right) + \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \frac{n}{4} \sigma_{ij} \sigma_{ik} (Z_i Z_j Z_k) \\
 &- \left[\sum_{i=1}^n Z_i \left(\sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n \frac{1}{4} \sigma_{ik} \sigma_{jl} \right) + \sum_{j=1}^n Z_j \left(\sum_{i=1}^n \sum_{k=1}^n \sum_{l=1}^n \frac{1}{4} \sigma_{ij} \sigma_{lk} \right) \right. \\
 &\quad + \sum_{i=1}^n \sum_{j=1}^n Z_i Z_j \left(\sum_{k=1}^n \sum_{l=1}^n \frac{1}{2} \sigma_{ik} \sigma_{jl} \right) \\
 &\quad + \sum_{i=1}^n \sum_{k=1}^n Z_i Z_k \left(\sum_{j=1}^n \sum_{l=1}^n \frac{1}{8} \sigma_{ik} \sigma_{jl} \right) + \sum_{j=1}^n \sum_{l=1}^n Z_j Z_l \left(\sum_{i=1}^n \sum_{k=1}^n \frac{1}{8} \sigma_{ik} \sigma_{jl} \right) \\
 &\quad + \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n Z_i Z_j Z_k \left(\sum_{l=1}^n \frac{1}{4} \sigma_{ik} \sigma_{jl} \right) + \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n Z_i Z_j Z_l \left(\sum_{k=1}^n \frac{1}{4} \sigma_{ik} \sigma_{jl} \right) \\
 &\quad \left. + \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n Z_i Z_j Z_k Z_l \left(\frac{1}{8} \sigma_{ik} \sigma_{jl} \right) \right] \\
 &+ A [\frac{1}{4} \sum_{i,j=1}^n (I + Z_i)(I + Z_j) - B \sum_{i=1}^n (I + Z_i)] \\
 &= \sum_{i=1}^n Z_i \left(\sum_{j=1}^n \sum_{k=1}^n \frac{n}{4} \sigma_{ij} \sigma_{ik} - \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n \frac{1}{4} \sigma_{ik} \sigma_{jl} + A * (\frac{n}{2} - B) \right) \\
 &+ \sum_{j=1}^n Z_j \left(\sum_{i=1}^n \sum_{k=1}^n \frac{n}{2} \sigma_{ij} \sigma_{ik} - \sum_{i=1}^n \sum_{k=1}^n \sum_{l=1}^n \frac{1}{4} \sigma_{ik} \sigma_{jl} \right) \\
 &+ \sum_{i=1}^n \sum_{j=1}^n Z_i Z_j \left(\sum_{k=1}^n \frac{n}{2} \sigma_{ik} \sigma_{jk} - \sum_{k=1}^n \sum_{l=1}^n \frac{1}{2} \sigma_{ik} \sigma_{jl} + \frac{1}{2} A \right) + \sum_{j=1}^n \sum_{k=1}^n Z_j Z_k \left(\sum_{i=1}^n \frac{n}{4} \sigma_{ij} \sigma_{ik} \right) \\
 &+ \sum_{i=1}^n \sum_{j=1}^n Z_i Z_k \left(- \sum_{j=1}^n \sum_{l=1}^n \frac{1}{8} \sigma_{ik} \sigma_{jl} \right) + \sum_{j=1}^n \sum_{l=1}^n Z_j Z_l \left(- \sum_{i=1}^n \sum_{k=1}^n \frac{1}{8} \sigma_{ik} \sigma_{jl} \right) \\
 &+ \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n Z_i Z_j Z_k \left(\frac{n}{4} \sigma_{ik} \sigma_{jk} \right) + \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n Z_i Z_j Z_k \left(- \sum_{l=1}^n \frac{1}{4} \sigma_{ik} \sigma_{jl} \right) \\
 &+ \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n Z_i Z_j Z_l \left(- \sum_{k=1}^n \frac{1}{4} \sigma_{ik} \sigma_{jl} \right) + \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n Z_i Z_j Z_k Z_l \left(- \frac{1}{8} \sigma_{ik} \sigma_{jl} \right)
 \end{aligned} \tag{A.1}$$

Appendix B. Covariance matrix of portfolio

The experiment utilize six groups of six randomly selected stocks, with the covariance matrices for each group presented as follows.

$$\Sigma_1 = \begin{bmatrix} 0.000392 & 0.000267 & 0.000109 & 0.000189 & 0.000106 & 0.000129 \\ 0.000267 & 0.000761 & 0.000152 & 0.000402 & 0.000188 & 0.000315 \\ 0.000109 & 0.000152 & 0.000756 & 0.000197 & 0.000170 & 0.000158 \\ 0.000189 & 0.000402 & 0.000197 & 0.000488 & 0.000169 & 0.000220 \\ 0.000106 & 0.000188 & 0.000170 & 0.000169 & 0.000484 & 0.000182 \\ 0.000129 & 0.000315 & 0.000158 & 0.000220 & 0.000182 & 0.000712 \end{bmatrix} \quad (\text{B.1})$$

$$\Sigma_2 = \begin{bmatrix} 0.000431 & 0.000103 & 0.000082 & 0.000121 & 0.000095 & 0.000070 \\ 0.000103 & 0.000790 & 0.000310 & 0.000184 & 0.000183 & 0.000232 \\ 0.000082 & 0.000310 & 0.000749 & 0.000127 & 0.000165 & 0.000159 \\ 0.000121 & 0.000184 & 0.000127 & 0.000598 & 0.000165 & 0.000239 \\ 0.000095 & 0.000183 & 0.000165 & 0.000165 & 0.000444 & 0.000044 \\ 0.000070 & 0.000232 & 0.000159 & 0.000239 & 0.000044 & 0.000966 \end{bmatrix} \quad (\text{B.2})$$

$$\Sigma_3 = \begin{bmatrix} 0.001012 & 0.000108 & -0.000095 & 0.000040 & 0.000167 & 0.000088 \\ 0.000108 & 0.000500 & 0.000255 & 0.000059 & 0.000027 & 0.000135 \\ -0.000095 & 0.000255 & 0.001169 & -0.000054 & -0.000050 & 0.000277 \\ 0.000040 & 0.000059 & -0.000054 & 0.000258 & 0.000024 & 0.000009 \\ 0.000167 & 0.000027 & -0.000050 & 0.000024 & 0.000319 & -0.000020 \\ 0.000088 & 0.000135 & 0.000277 & 0.000009 & -0.000020 & 0.000969 \end{bmatrix} \quad (\text{B.3})$$

$$\Sigma_4 = \begin{bmatrix} 0.000259 & 0.000211 & 0.000177 & 0.000197 & 0.000131 & 0.000110 \\ 0.000211 & 0.000591 & 0.000295 & 0.000272 & 0.000312 & 0.000257 \\ 0.000177 & 0.000295 & 0.001006 & 0.000132 & 0.000146 & 0.000308 \\ 0.000197 & 0.000272 & 0.000132 & 0.001350 & 0.000197 & 0.000126 \\ 0.000131 & 0.000312 & 0.000146 & 0.000197 & 0.000288 & 0.000154 \\ 0.000110 & 0.000257 & 0.000308 & 0.000126 & 0.000154 & 0.001190 \end{bmatrix} \quad (\text{B.4})$$

$$\Sigma_5 = \begin{bmatrix} 0.001114 & 0.000070 & 0.000047 & 0.000200 & 0.000132 & 0.000025 \\ 0.000070 & 0.000743 & 0.000113 & 0.000067 & -0.000001 & 0.000008 \\ 0.000047 & 0.000113 & 0.000465 & 0.000172 & 0.000129 & 0.000297 \\ 0.000200 & 0.000067 & 0.000172 & 0.000549 & 0.000159 & 0.000181 \\ 0.000132 & -0.000001 & 0.000129 & 0.000159 & 0.000511 & 0.000129 \\ 0.000025 & 0.000008 & 0.000297 & 0.000181 & 0.000129 & 0.001768 \end{bmatrix} \quad (\text{B.5})$$

$$\Sigma_6 = \begin{bmatrix} 0.000116 & 0.000017 & 0.000106 & 0.000169 & 0.000061 & 0.000032 \\ 0.000017 & 0.001212 & 0.000102 & 0.000082 & 0.000060 & 0.000177 \\ 0.000106 & 0.000102 & 0.000346 & 0.000194 & 0.000125 & 0.000205 \\ 0.000169 & 0.000082 & 0.000194 & 0.000436 & 0.000121 & 0.000099 \\ 0.000061 & 0.000060 & 0.000125 & 0.000121 & 0.000274 & 0.000160 \\ 0.000032 & 0.000177 & 0.000205 & 0.000099 & 0.000160 & 0.000696 \end{bmatrix} \quad (\text{B.6})$$

Appendix C. Definition and hamiltonian derivation of MV model

From Eq. (1), the MV model is defined as follows:

$$\begin{aligned} \min_{w_i} & q \sum_{i,j=1}^n x_i x_j \sigma_{ij} - (1-q) \sum_{i=1}^n x_i \mu_i \\ \text{s.t.} & 1^T X = B \end{aligned} \quad (\text{C.1})$$

where $\Sigma = (\sigma_{ij}) \in R^{n \times n}$ is the covariance matrix of the investment portfolio, μ_i represents the expected return of each investment product, $X = (x_1, x_2, \dots, x_n)$, B represents the number of investment assets.

The MV model for portfolio optimization problem can be translated into an objective function as follows:

$$F_{MV} = q \sum_{i,j=1}^n x_i x_j \sigma_{ij} - (1-q) \sum_{i=1}^n x_i \mu_i + A \left(\sum_{i=1}^n x_i - B \right)^2 \quad (\text{C.2})$$

where $A \left(\sum_{i=1}^n x_i - B \right)^2$ is a constraint term, A is the penalty coefficient.

When mapped to the problem Hamiltonian H_p by $x_i = \frac{1+z_i}{2}$, it can be expressed as Eq. (C.3), and the detailed derivation is shown in Eq. (C.4).

$$H_p = Z_i Z_j \sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{1}{2} q \sigma_{ij} + \frac{1}{2} A \quad (\text{C.3})$$

$$\begin{aligned} & + \sum_{i=1}^n Z_i \left(\sum_{j=1}^n \frac{1}{2} q \sigma_{ij} + A * \left(\frac{n}{2} - B \right) + \frac{1}{2} (q-1) \mu_i \right) \\ \hat{F}_{MV} & = q \sum_{i,j=1}^n x_i x_j \sigma_{ij} - (1-q) \sum_{i=1}^n x_i \mu_i + A \left(\sum_{i=1}^n x_i - B \right)^2 \\ & = q \sum_{i,j=1}^n \left(\frac{1+z_i}{2} \right) \left(\frac{1+z_j}{2} \right) \sigma_{ij} - (1-q) \sum_{i=1}^n \left(\frac{1+z_i}{2} \right) \mu_i + A \left(\sum_{i=1}^n \left(\frac{1+z_i}{2} \right) - B \right)^2 \\ & = \frac{q}{4} \sum_{i,j=1}^n (\sigma_{ij} + z_i \sigma_{ij} + z_j \sigma_{ij} + z_i z_j \sigma_{ij}) - \frac{(1-q)}{2} \sum_{i=1}^n (\mu_i + z_i \mu_i) \\ & \quad + A \left(\frac{n}{2} + \frac{1}{2} \sum_{i=1}^n z_i - B \right)^2 \end{aligned} \quad (\text{C.4})$$

Data availability

All data and codes that support the findings of this study are available at <https://github.com/wqs1999/QAOA-Portfolio>.

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